Module 12 — Role-specific fresher mock interviews

These mocks are intentionally different from the concept exercises. They test whether you can combine skills under time pressure.

Mock A — Data Analyst / Product Analyst (90 minutes, 100 marks)

1. SQL analysis — 25 marks, 25 min: identify customers whose second purchase occurred within 60 days; calculate repeat rate by signup cohort.

2. Excel/BI — 15 marks, 15 min: propose a report for subscription renewals and explain the model.

3. Statistics — 15 marks, 15 min: interpret a test result with effect size and confidence interval.

4. Product case — 25 marks, 25 min: free-trial activation falls 8 percentage points.

5. Communication — 20 marks, 10 min: present finding, evidence, recommendation and caveat.

Answer expectations

• SQL must define cohort, completed purchase and date window.

• BI must define fact grain, dimensions, measures and reconciliation.

• Statistics must not confuse p-value with probability the null is true.

• Product case must validate instrumentation, decompose funnel and segment.

• Communication must distinguish fact from hypothesis.

Mock B — Junior Data Engineer (100 minutes, 100 marks)

1. SQL — 25 marks, 25 min: latest inventory state per SKU/store; detect source duplicates.

2. Python — 20 marks, 20 min: parse nested JSON, validate required fields, quarantine invalid records.

3. DBMS — 15 marks, 15 min: transaction, index and isolation scenario.

4. Pipeline design — 25 marks, 25 min: daily supplier feed with corrections and missing files.

5. Reliability/behavioural — 15 marks, 15 min: explain a failed run, retry and communication.

Answer expectations

• Deterministic latest-row selection.

• Python separates parsing, validation and persistence.

• DBMS answer includes engine-specific caveat.

• Design includes raw retention, idempotency, late corrections, backfill, alerting and SLA.

• Behavioural response states actual contribution.

Mock C — Mixed entry-level data role (60 minutes, 60 marks)

• 15 min SQL: rank sellers by region.

• 10 min Python: frequency and deduplication.

• 10 min DBMS: explain ACID and index trade-offs.

• 15 min business case: revenue flat but active buyers down.

• 10 min project defence: architecture and one limitation.

Marking guide

Award full marks for correct result plus assumptions, edge cases and explanation. Award partial marks for a correct approach with a small syntax issue if the candidate can debug it. Deduct for unsupported assumptions, silent data loss, incorrect metric denominator or fabricated project claims.

Interview communication framework

For coding: clarify → state approach → code → test → complexity.

For analytics: define metric → validate data → decompose → quantify → recommend.

For design: requirements → architecture → data model → failure handling → monitoring/security → trade-offs.

For behavioural: STAR, with specific personal actions and honest outcomes.

External timed practice for this module

Analyst / Product Analyst track

• DataLemur: complete 10 mixed SQL questions tagged easy/medium, prioritising joins, aggregation, ranking and product metrics.

• StrataScratch: complete 5 business-oriented SQL questions; explain assumptions before coding.

• LeetCode SQL 50: complete the full plan, then reattempt failed questions under a 45-minute limit.

• Kaggle Learn: complete Python and Pandas exercises; perform one independent dataset analysis.

Junior Data Engineer track

• HackerRank SQL: complete 10 mixed SQL problems.

• LeetCode: solve 10 Easy and 3 Medium array/hash-map problems.

• Exercism Python: complete 5 exercises and review edge cases.

• DataTalksClub Zoomcamp: complete one selected ingestion or warehouse lab and explain the architecture.

Final rule

Do not count a problem as completed merely because the platform accepted the code. Re-solve it from a blank editor after 3–7 days and explain complexity, grain, assumptions and edge cases aloud.